DRY LAND AGRICULTURE

Last 5 year previous year question 2019-2023

1. **Dry land farming is practiced in the region of having annual rainfall of (2021)**
   a. Below 750 mm
   b. 750 mm to 1150 mm
   b. Above 1150 mm
   d. None of the above

2. **Agriculture contingency plan was formulated by the government and the ICAR for the better cropping practices and benefiting the farmers of which region (2019)**
   a. Dry land regions
   b. Rainfed regions
   c. Himalayan region
   d. All of the above

3. **Rainfed agriculture is practiced in regions of receiving __________ rainfall annually. (2022)**
   a. 750 mm
   b. More the 800 mm
   c. More the 1150 mm
   d. None of the above

4. **What is the suction pressure for water absorption by the plants during permanent wilting point. (2021)**
   a. -15 bar
   b. 15 bars
   c. 0 to -15 bar
   d. Above 15 bars

5. **What is the gross irrigated area in India which is having irrigation facility (2020)**
   a. 143.8
   b. 42
   c. 43.8
   d. 44.8

6. Salinity of the soil is a major problem of soil in which of the following region (2023)

a. Rainfed regions
b. Warm & humid regions
c. Himalayan regions
d. Arid & semi-arid regions

7. Mulches are used to reduce ________ loss from the soil. (2019)

a. Photosynthesis
b. Water loss by transpiration
c. Water loss by evaporation from soil
d. All of the above

Answers to the question

1. B 2.A 3.C 4.A 5.C 6.D 7.C

DRY LAND AGRICULTURE

Crops grown entirely under rainfed condition is called as dryland agriculture.

Dry land agriculture is grouped into three categories on the basis of amount of rainfall received in those regions.

DRY LAND AGRICULTURE

Dry farming | **Dryland farming** | **Rainfed farming**

Flow Chart I- Types of Dryland Agriculture

1. Dry Farming

Cultivation of crops where rainfall is **less than 750 mm per annum** is called as a dry farming. Dry farming regions are arid regions of the country

Main features of this region

i. Prolonged **dry spells are very common** during crop period.
ii. More **frequent crop failures**.
iii. **High Evo transpiration** (moisture loss)
iv. Very low **moisture presents in soil**.
v. **Less socio-economic** development.
vi. **High saline soil**
vii. **Cropping period is highly** dependent on the rainfall

Measures to be adopted- soil & moisture conservation practice need to be adopted

Major regions of dry farming in India – western Rajasthan, Bundelkhand region, Gujarat & all the arid regions.

2. Dryland Farming

Cultivation of crops in area receiving rainfall **above 750-1150 mm per annum** is called as dryland farming. Dryland farming is practiced in all the semiarid regions of the country.

Image (i)- Jowar Is Grown in A Dryland Region

Main Characteristics of This Region

i. **Dry spell occurs** during crop period
ii. **Less frequent** crop failure.
iii. **Salinity of soil**

Major regions where dryland farming is practiced- Rajasthan, Madhya Pradesh, Gujarat, Tamil nandu, Andhra Pradesh and all the semi-arid regions of the country.

3. Rainfed Farming

Cultivation of crops in regions receiving **more than 1150 mm annually** is known as rainfed farming. Rainfed farming is practiced in humid regions of the country or regions receiving high rainfall.

Image (ii)- paddy is transplantation in rainfed condition

Main features of this region

i. Crop failure is very rare
ii. Very rare or no dry spell
iii. Waterlogging/ drainage is major problem
iv. Acidic soil is a constrain

Major region- coastal areas, north eastern states etc.

Importance Of Dryland in India

143.8-million-hectare land is cultivated in India, out of which the **irrigated area is 30.5 percent (majorly in Gangetic plains) and 69.5%** is under dryland or rainfed area, almost all the coarse grain, **about 75% pulses and oilseed, 2/3 of mustard**, most of soybean and groundnut are grown under rainfed condition.

Table 1. Difference Between Dryland & Rainfed Farming

CONSTITUENT	DRYLAND FARMING	RAINFED FARMING
Rainfall Annually (mm)	Less than 800mm	More than 800 mm
Moisture Availability to The Crop	Shortage	Enough
Growing season (days)	Less then 200	More then 200
Growing regions	Arid, semi-arid, as well as uplands of subhumid & humid region	Humid & subhumid region
Cropping system	Single crop or intercropping	Intercropping or double cropping
Constrains	Wind & water erosion	Water erosion

Problems & Constrains in Dryland Agriculture

1. **Less or lack of proper moisture** for cultivating crops is the major constrain in dryland agriculture.
2. **High temperature** with Very low or erratic rainfall.
3. **Low water table** so very less irrigation facility like tubewell, canals etc.
4. **Degraded soil** due to very less organic matter in the soil
5. **Less fertile soil,** less nutrient present in soil.
6. **High PET** (potential evo transpiration rate) means high water
7. **Loss of water from transpiration** by plants & evaporation from soil.
8. **Only one crop** or intercrops are grown in this region in one season only.
9. **Very less crop intensity** and no crop diversification
10. **Salinity in the soil** is a very common problem
11. **Crop failure** is common due to frequent dry spells
12. **Agriculture is more of subsistence** in nature rather than profitable.

13. Low socio-economic development

***The main constrain in dry land agriculture is less rainfall which causes very less moisture in this region. This region is very prone to drought. Which directly or indirectly impact the other factors in this region like soil fertility, degraded soil, less socio-economic development etc**

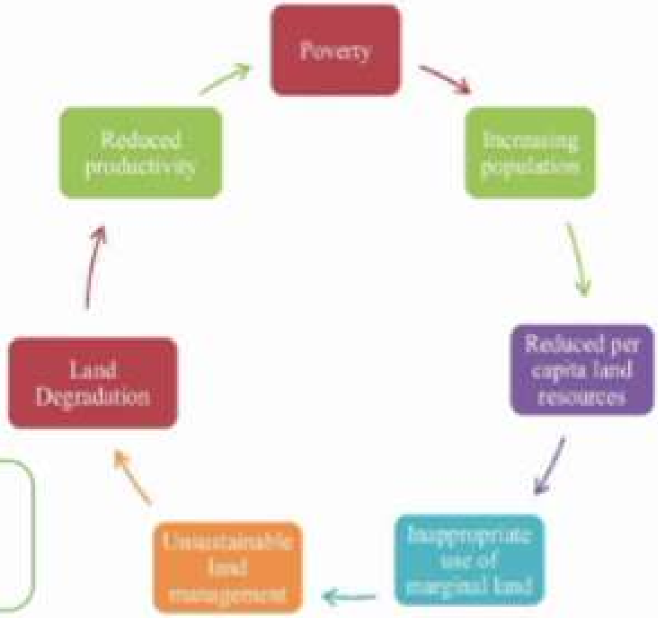
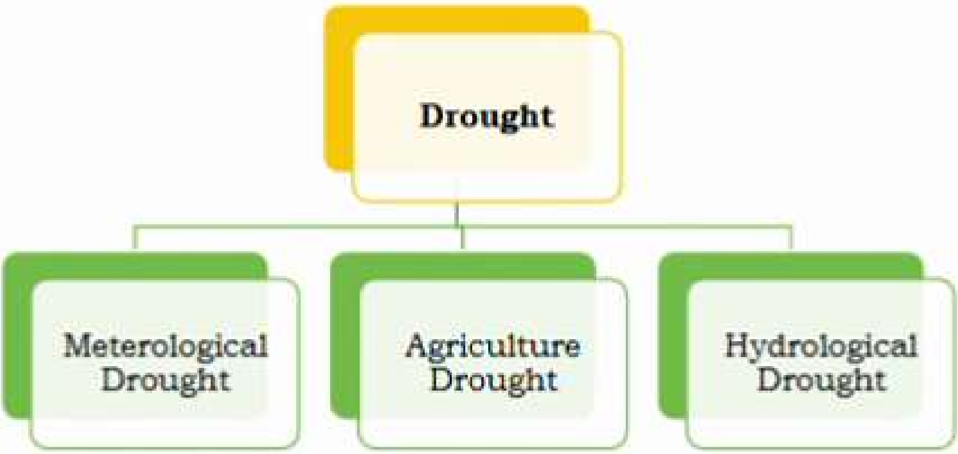


Image (iii)-reasons of low socio-economic development in dryland regions

DROUGHT AND MOISTURE STRESS

Drought- An extended **period of deficient rainfall compared to normal rainfall** of the region is called as drought

Drought Is Classified Into -Meteorological Drought, Agriculture Drought & Hydrological Drought.



Flow Chart 2 – Types Of Droughts

Meteorological Drought-

It refers to the substantial deficit or rainfall relative to the average of that region. According to the IMD meteorological drought is a condition **when there is 25% decrease in the rainfall** for a given period of time in a region.

Agriculture drought-

It refers to the extended dry period in which lack of rainfall results in **insufficient moisture in the root zone** of the soil causing adverse effects on crop.

Hydrological drought –

It is **extended dry period leading to marked depletion** of the surface water and consequent **drying up of reservoirs, lakes, streams, rivers, cessation** of spring flows and fall in ground water.

Other related terms

- **Dry spell** – rainless **period more than 10 days** in light soil (sandy) areas and 15 days in heavy soil (clay) areas.
- **Drought**- it is prolonged dry spell resulting wilting or drying of crops
- **Desertification**- it is the land degradation in arid, semi-arid, and dry humid areas resulting from factors like climatical and human activities

Low Moisture Stress In Dryland Agriculture

Low moisture stress impact various physiological functioning in the plants.

Development of moisture stress in crops – Water **deficit in plants occur when transpiration exceeds the rate of absorption of water by the plants.** It can be caused by excess of water loss, less absorption or by both.

Impact Of Low Moisture Stress on Plants

1. **Incipient wilting or midday depression-** when after irrigation or rainfall transpiration exceeds the rate of absorption of water from soil in the noon during hot days/ summers. This type of water deficit occurs for a very short duration
2. **Permanent wilting point** – when the soil **moisture reaches to -15 bar**, plants show wilting symptom during the day but not die.
3. **Ultimate wilting point**- when the soil **moisture reaches -6 MPa or -60 bar** plant will die due to moisture stress.

Moisture Conservation in Dryland Agriculture

Soil Moisture- The amount of **water present in the top soil at a given point** of time is called as a soil moisture

Soil moisture can be influenced by the following

- Rainfall
- Soil texture
- Soil depth

Methods Of Conserving Soil Moisture In Dryland Agriculture

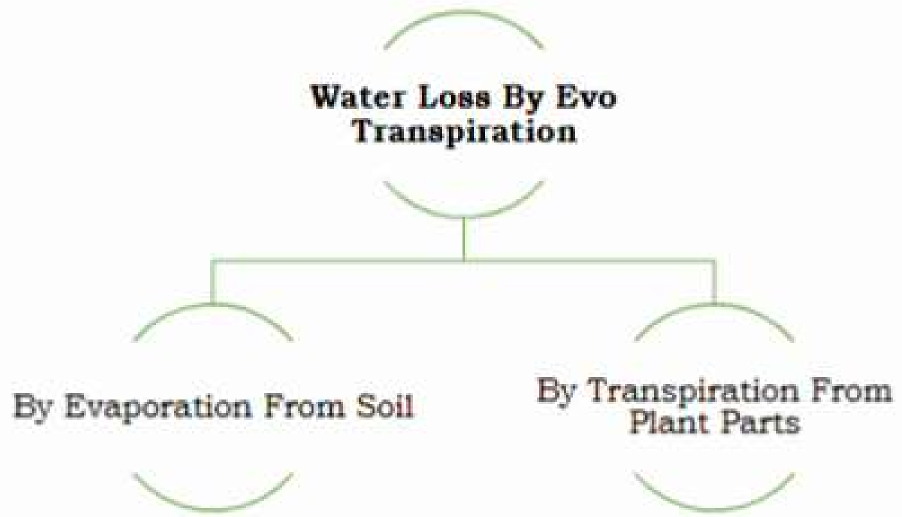
1. Soil moisture storage

- **Reducing runoff**- water runoff increases **when there is high intensity of rainfall due to low infiltration, so more opportunity time** should be provided to the runoff water to increase the rate of infiltration.
- **Application of farm yard manure increase the soil moisture by increasing** soil water holding capacity.
- **Disc ploughing during summer** and contour bunding enhances the soil moisture storage in the soil.
- **Increase rate of infiltration** by maintaining mulches and addition of crop residue in fields.

2. Tillage

- Tillage increases the **water holding capacity, reduces runoff, increases infiltration** & conserve soil moisture.
- **Subsoiling is the best practice** to conserve soil moisture and gives best result during fallow period and hence more water in soil is present for winter crops.
- Contour cultivation or **cultivation along the slope increase soil moisture**.
- **Making conservation furrows or opening furrows at 3-4 m** interval between crop row reduce moisture loss.
- **Tied ridges are also beneficial** in conserving soil moisture.

3. Reduction In Evo-Transpiration (ET) Losses



Flow Chart 3- Types Of Water Loss from A Agri Field

Eva-Transpiration Loss Of Water Can Be Prevented By

1. Wind breaks or shelter belts
2. Mulching
3. Anti-transpirant
4. Weed control

Wind break / shelter belt –Evo transpiration loss increases during windy condition

Wind Breaks – wind breaks **are any structure artificial or natural** which reduce & obstruct the wind speed.

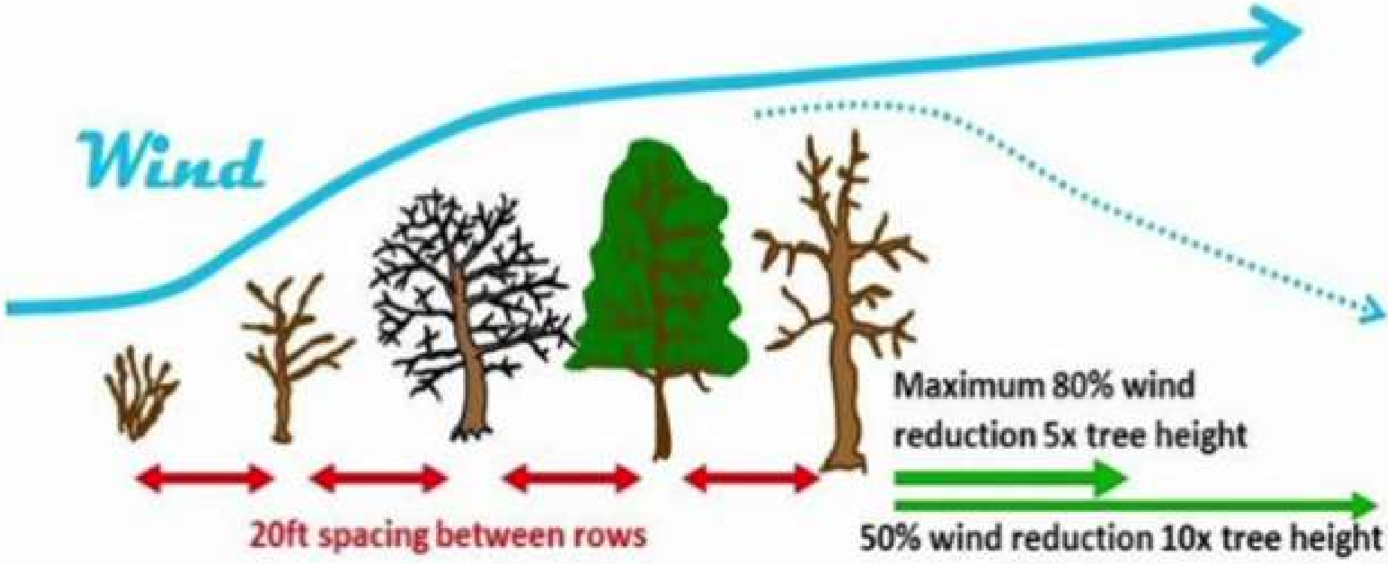


Image (iv)- Functioning of wind break

Shelter Belt- a long row of **plants planted across the direction of wind**. They do not obstruct the wind speed completely

Image (v)- Eucalyptus shelter belt planted across wheat field

a. The reduction in **wind speed reduce evaporation loss & water** is made available to plants.
b. They also **reduce wind erosion**

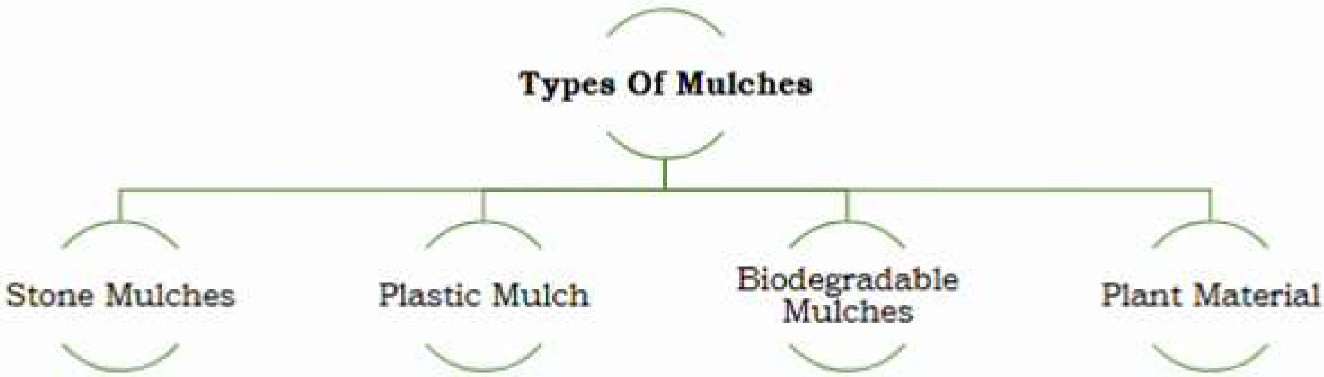
Plants used as shelter belt – casuarina, subabul, bamboo, phalsa, etc.

1. Mulches –

Mulch – mulch is any material applied on the soil surface to check evaporation and increase soil moisture. **Approximately 60 to 75% water is** lost through evaporation and this loss can be prevented to an extent by application of mulches.

Other benefit of mulches

- Soil conservation
- Moderation of temperature
- Reduction in soil salinity
- Weed control
- Improvement in soil structure



Flow chart 4- types of mulches

a. Stone mulch-

Application of sizable stones in a field **offers potential benefit for conservation of moisture in arid and semiarid regions by reducing evaporation**. This will help in increase in yield also.

Image (vi)- stone mulching

b. Plastic mulch

Plastic **materials like polyethylene, polyvinyl chloride** is also used as mulching materials. **Plastic mulching increase soil temperature** there by increase the plant growth and development along with reduction in loss of soil moisture.

Image (Vii)- Plastic Mulch In Strawberry Field

c. Biodegradable mulches-

Polymers like **poly lactic acid, poly butylene adipate coterephthalate, starch-based** polymer **is used as a biopolymer and offer very less impact on the environment**. These films can degrade easily when exposed to bioactive environments like soil, water etc.

Image (viii)-biodegradable mulching paper

d. Plant material –

Plant material like **crop residue, straw, groundnut shells, etc act as a very good mulch material** without any impact on environment. These materials also add organic material in the soil.

Image (ix)-Straw Mulching in Cauliflower Field

2. Antitranspirants

Antitranspirants is any material applied **to the transpiring plant surface like leaves**, stems **etc to reduce the water loss from plants.**

Nearly 99% of water absorbed by the plants is lost by absorption.

Types Of Antitranspirants

- Stomata Closing
- Film Forming
- Reflectant Type
- Growth Retardants

Flow Chart (V)-Types Of Anti- Transpirants

a. Stomata Closing-

Stomata closing anti-transpirant function by closing the stomatal pores present on the leaves surface from which maximum transpiration occurs

Examples -phenyl mercuric acetate and atrazine

b. Film Forming Type-

The mode of action of film forming antitranspirant **is that they form a thin waxy layer on the leaf surface and** retard the escape of water from the leaves by formation of a physical barrier.

Examples- mobileaf, hexadeconol, silicone etc.

c. Reflectant Type-

These are white materials which form a **coating on the leaves and increase the leaf reflectance (albedo)**. By reflecting the radiation, they reduce leaf temperatures and vapour pressure gradient from leaf to atmosphere and thus reduce transpiration.

Examples- 5 per cent kaolin spray, diatomaceous earth product (celite) etc

d. Growth Retardants –

These chemicals reduce shoot **growth and increase root growth and thus enable the plants to resist drought**. They may also, induce stomatal closure.

Example- Cycocel is one such chemical useful for improving water status of the plant.

Disadvantage Of Antitranspirants-

Antitranspirants generally reduce photosynthesis. Therefore, their **use is limited to save the crop from death under severe moisture stress**. If crop survives, it can utilise the rainfall that is received subsequently.

Advantage Of Antitranspirants –

Antitranspirants are also **useful for reducing the transplantation shock of nursery plants**. They have some practical use in nurseries and horticultural crops.

3. Weed Control-

- ✓ Prompt **weed control eliminates the competition of weeds** with crops for limited soil moisture.
- ✓ **Transpiration rate from weeds is** more compared to crops.
- ✓ **Effective weed control in dryland agriculture** leads to increasing availability of soil moisture to crops.
- ✓ This is the **most useful measure to reduce transpiration losses**.

Other method

- ➢ **Spraying nutrient solution**-Nutrient solution spray is recommended in the event of revival of rain and release of moisture stress.

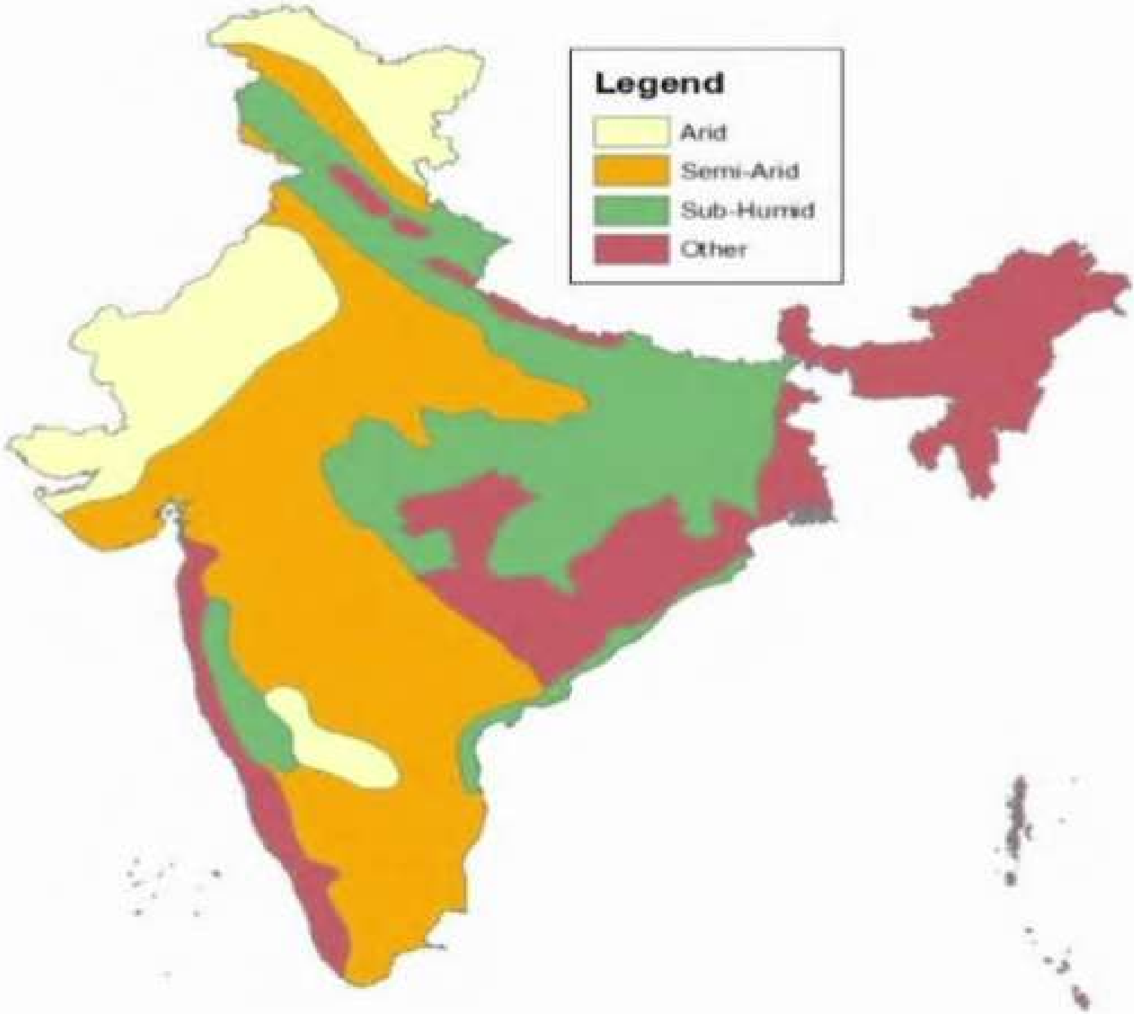
- **Urea or DAP spray (2% solution)** is useful for quicker regeneration of crops like legumes and castor after rain.

Distribution of dryland farming in India

Our country has **fertile cultivable land and receives the highest rainfall** on per unit area basis anywhere in the world due to short duration of rainfall in a year.

✓ **128 districts** in India **have been recognized as dryland farming** areas in our country.

✓ **Out of the 128 districts 91 districts are spread in the states of Madhya Pradesh, Chhattisgarh, Uttar Pradesh and Tamil Nadu**, representing typical dry farming tracts. Rest of the districts belongs to Central Rajasthan, Saurashtra region of Gujarat and rain shadow region of the Western Ghats.



Map 1- distribution of dryland in India

Table 2. Area under irrigated and rainfed conditions in India

Class	Area (m ha)
Total arable	143.8
Dryland	34.5
Rainfed	65.5
Irrigated	43.8

Major Crops Cultivated in Dryland Conditions

- ✓ **Millets**- such as bajra, ragi, sorghum etc
- ✓ **Oilseeds**- mustard, rapeseed, groundnut etc
- ✓ **Pulse crops**- pigeon pea, gram and lentil, soyabean, moong, chick pea etc
- ✓ **Almost 80% of maize and Jowar, 90 per cent of Bajra** and approximately 95% of pulses and 75% of oilseeds are obtained from dryland agriculture.
- ✓ In addition to **these, 70% of cotton is produced through dryland agriculture**. Dryland areas also contribute significantly to wheat and rice production. **33% of wheat and 66% of rice are still rainfed.**

Government Initiatives for Dryland Agriculture

1. **The National Mission for Sustainable Agriculture (NMSA)**, which is one of the eight missions under the **National Action Plan on Climate Change (NAPCC)** seeks to address issues associated with climate change.

2. **Sub-Mission on Agroforestry** under the framework of National Mission for Sustainable Agriculture (NMSA) **has been launched during 2016-17 for a period of 4 years (2016-17 to 2019-20).**

3. **The Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)** was launched on 1st July, 2015 with the **motto of 'Har Khet Ko Paani' for providing end-to end solutions in irrigation supply chain, viz. water sources, distribution** network and farm level applications.

4. **Agriculture Contingency Plan:** CRIDA, ICAR has prepared district **level Agriculture Contingency Plans in collaboration with state agricultural universities** using a standard template to tackle aberrant monsoon situations leading to drought and floods, extreme events adversely affecting crops, livestock and fisheries.

5. **National Watershed Development Project for Rainfed Areas (NWDPRA)**- The scheme of **National Watershed Development Project for Rainfed Areas (NWDPRA) was launched in 1990-91 based** on twin concepts of integrated watershed management and sustainable farming systems.

6. **Rainfed Area Development Programme (RADP)**- Rainfed Area Development Programme (RADP) was **implemented as a sub-scheme under Rashtriya Krishi Vikas Yojana (RKVY).**